Professional Education



Applied Agentic AI for Organizational Transformation

Leverage AI Agents to Elevate Efficiency and Innovate Models



AI is no longer just a tool—it's a strategic imperative. Leaders who understand how to harness Generative and Agentic AI will shape the future of business, not just react to it.

Dr. Abel Sanchez

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AI Transformation: Leading With Generative and Agentic AI



In a world rapidly reshaped by artificial intelligence, strategic leadership demands more than awareness, it requires action. The Applied Agentic AI for Organizational Transformation course from MIT Professional Education equips decision-makers with the insights, tools, and frameworks to lead successful AI-driven transformation across their organizations. In addition to focusing on prominent technologies, the program also dives deep into generative and agentic AI, empowering leaders to drive practical innovation, navigate ethical and governance considerations, and create measurable business value.

Designed for mid-to-senior working professionals in strategic roles, this 8-week course is your opportunity to explore the AI landscape, from foundational concepts and tools like ChatGPT to enterprise integration, governance, compliance, and real-world adoption. Weekly hands-on mini-projects give you a safe, practical environment to experiment with AI applications, including AI-powered workflows and business use cases.

At the end of the course, you will develop a structured AI adoption plan for your organization, applying course concepts to identify opportunities for operational improvement, innovation, and business transformation. This capstone project is supported by hands-on assessments and expert guidance from MIT instructors.

Professional Education

Delivered in collaboration with global alumni -excellence or nothing-

Why Elevate Your Digital Strategy?

$100 trillion

will be added to the world economy through digital transformation by 2025.

Source: World Economic Forum

79%

of executives believe AI will significantly transform their industry in the next three years.

Source: PwC Global AI Study, 2023

76%

of executives agree that organizations need to drastically redesign the experiences that unite technology and people in more human-centered ways.

Source: Accenture

Why This Course?



- Understand the building blocks of generative and agentic AI, including how they differ, how they work, and what business functions they can transform.
- Design AI-integrated digital strategies using cloud, APIs, enterprise systems, and agent-based workflows to deploy real-world solutions.
- Apply AI through mini-projects and gain hands-on experience developing AI-powered workflows, creating digital solutions, and evaluating AI use cases in business contexts.
- Plan for ethical considerations, and legal compliance, as well as addressing risks like deepfakes, misinformation, and prompt injection with sound governance strategies aligned with emerging regulatory frameworks such as GDPR, CCPA, and HIPAA.

8 weeks | 8–10 hours a week | Online

Learning Outcomes



1. Understand the fundamentals of generative and agentic AI, including their architectures, use cases, and future potential.
2. Assess where and how AI can improve business outcomes, from operational workflows to customer engagement, while evaluating the viability and cost-effectiveness of AI adoption.
3. Integrate AI systems into existing digital ecosystems, leveraging cloud infrastructure, APIs, and enterprise platforms.
4. Evaluate AI platforms for strategic fit, capabilities, business value, and risk.
5. Develop agent-based solutions to orchestrate multistep workflows and enhance automation.
6. Navigate governance and compliance challenges, including ethical deployment and regulatory frameworks like GDPR, HIPAA, and CCPA.
7. Build a strategic AI roadmap, culminating in an actionable adoption plan or executive pitch, supported by change management considerations for successful implementation.

Key Takeaways

01 **Lead with Confidence in the Age of AI:** Understand the strategic implications of generative and agentic AI, and how to integrate them into your business models, customer journeys, and operations.

02 **Understand the Tools of Transformation:** Gain hands-on exposure to industry-leading platforms such as ChatGPT and Ollama, learning how to evaluate, prototype, and deploy AI-powered solutions while assessing implementation feasibility and cost.

03 **Drive Responsible Innovation:** Develop a clear understanding of AI ethics, bias mitigation, compliance frameworks (like GDPR, CCPA, and HIPAA), and governance best practices.

04 **Communicate AI Value Across the Organization:** Learn to articulate AI's business potential in clear, actionable language that engages stakeholders across functions.

05 **Deliver a Strategic AI Roadmap:** Walk away with an executive-ready AI adoption plan or strategic presentation tailored to your organization's needs.



Who Is This Course For?

This course is designed for decision-makers who recognize the strategic urgency of AI and want to lead its adoption effectively within their organizations. It is tailored for professionals who may not build AI tools themselves—but are responsible for choosing, funding, governing, or operationalizing them.

This course is ideal for:

C-suite executives (CEOs, CIOs, CTOs, CMOs, COOs) looking to make informed, forward-thinking decisions about AI integration.

Business leaders and functional heads driving digital innovation in areas like operations, strategy, marketing, or product.

Managers and team leads responsible for modernizing workflows and aligning teams with technology change.

Technical professionals looking to bridge the gap between engineering and leadership in digital transformation efforts.

Consultants and advisors helping clients navigate AI-powered change in complex organizational contexts.

This course assumes no prior experience with AI tools or programming. However, a strategic mindset and openness to hands-on experimentation will allow participants to get the most out of the experience.

Instructor



Dr. Abel Sanchez
Research Scientist, MIT

Dr. Sanchez is the architect of the global network “The Internet of Things” and data analysis platforms for SAP, Ford, Johnson & Johnson, Accenture, Shell, Exxon Mobil and Altria. In cyber security, he has developed cyber-attack impact analyses for the U.S. Department of Defense and a password firewall for IARPA. He holds a PhD from MIT and teaches cyber security, engineering, blockchain and data science. He has contributed to educational software for Microsoft and the Accenture Technology Academy. He has produced over 150 educational videos and has 10 years of experience with LMS across the Americas, Asia and Europe.

Instructor



Prof. John R. Williams

Professor, MIT Department of Civil and Environmental Engineering; Affiliated Faculty, MIT Center for Computational Science and Engineering

John R. Williams is a globally recognized leader in **AI-driven systems, high-performance computing, and energy infrastructure,** with a career at the intersection of **advanced computation and the energy sector**. His work has consistently focused on enabling **large-scale, mission-critical energy systems**—from subsurface modeling to modern AI-enabled infrastructure.

He is co-inventor of the **Discrete Element Method (DEM)**, a foundational simulation capability used extensively across **oil & gas, mining, and energy-intensive industries** to model granular and discontinuous materials. DEM remains a core technology for understanding subsurface behavior, drilling dynamics, and material flows at inustrial scale.

Prior to academia, Dr. Williams served as **Vice President of Applied Mechanics**, where he led consulting engagements for the oil and gas sector, applying advanced numerical methods and simulation to real-world energy challenges. This industry grounding has shaped a career in which **all research is explicitly aligned with energy systems, infrastructure resilience, and large-scale industrial operations**.

His work now directly addresses one of the most critical constraints in the U.S. economy: **the need to rapidly scale energy supply to support AI-driven data center expansion and gigawatt-class infrastructure.**

Dr. Williams' expertise spans **AI, agentic systems, and high-performance distributed computing**, with a focus on:

- **Energy-aware computation** and optimization of data center workloads
- **Grid-integrated AI systems** for real-time control and resilience
- **Latency- and fault-tolerant architectures** for energy-critical operations

His innovation in **fault-tolerant computing via container-level migration**—enabling seamless recovery without code modification—received the **2019 IEEE High Performance Extreme Computing Award for Best Innovation**, with direct relevance to **high-availability energy and compute infrastructure.**

At Massachusetts Institute of Technology, he has led programs that connect engineering, computation, and large-scale systems deployment:

- **Co-Director, Systems Design and Management (SDM) Program** — training leaders to architect complex energy and infrastructure systems
- Director, **Auto-ID Laboratory** (2006–2012), where foundational work on the **Internet of Things (IoT)** enabled large-scale sensing and control—now critical for modern energy systems

He has authored **250+ publications**, multiple books, and contributed to national strategy, including the UK government's **Future of Manufacturing Foresight Report**.

Alongside Bill Gates and Larry Ellison, he was named one of the 50 most powerful people in computer networks, reflecting the global impact of his work on distributed and infrastructure-scale systems.

He delivered the **introductory two-day executive workshop on "Energy and AI" at CERAWeek 2026**, engaging senior leaders on how AI is reshaping **energy production, grid operations, and data center demand.**

Energy-Centric Industry Impact

Dr. Williams has advised leading global energy companies and institutions, including:

- **Shell, ExxonMobil, Total, ARAMCO, Schlumberger**
- **National laboratories and defense agencies** (Sandia, Lincoln Laboratory, IARPA)

His work focuses on deploying computation and AI into operational energy systems, including:

- Subsurface and materials modeling for extraction and production
- Smart grid and energy market systems (e.g., SAP smart grid architectures)
- Energy-intensive supply chain and industrial system simulation
- Cybersecurity and resilience for critical infrastructure

Strategic Relevance: Energy × AI × Infrastructure

Dr. Williams' current work is directly aligned with the structural shift underway:
AI is no longer just a compute problem—it is an energy systems problem

He is focused on enabling:

- Co-design of data centers and power systems (compute + generation + grid integration)
- AI-driven optimization of gigawatt-scale energy assets
- Agentic AI for autonomous energy system operation and control
- Resilient architectures for always-on infrastructure at national scale

Current Focus: Agentic AI for Energy Systems

His work positions Agentic AI as core infrastructure for the energy transition:

- Multi-agent systems for real-time grid and energy asset management
- Integration of AI into physical energy systems and industrial operations
- Design of principled, scalable AI architectures for high-consequence environments

He teaches at MIT:

- Architecting Software Systems
- Engineering Computation and Data Science

and delivers executive programs in **Applied Generative AI and Agentic Computing,** with a strong emphasis on **energy and infrastructure applications.**

Industry Positioning

- **Energy-first systems architect:** Career-long focus on oil & gas, energy systems, and infrastructure
- **AI × Energy convergence leader:** Deep expertise in linking compute demand with power generation and grid systems
- **Builder of mission-critical systems:** Proven delivery in high-reliability, high-scale environments
- **Trusted advisor:** Global energy companies, national labs, and government agencies

Course Outline

Module 1
Foundations of Generative and Agentic AI

1. Evaluate the strategic value of AI functionalities such as chatbots, reasoning, and multimedia
2. Construct an evaluation of the cost of an AI system
3. Distinguish between major AI model types and terminology

Module 2
The Rise of Agentic AI and Emerging Platforms

1. Explain the most relevant AI platform or approach for a specific sector and its application to agentic AI use cases
2. Evaluate the key factors influencing the selection of open-source versus proprietary AI platforms within a specific organizational context
3. Develop a Landing Page Using AI
4. Prompt AI to create a visual mock-up and functional HTML code
5. Activate the code by saving and reuploading
6. Explain a new AI workflow in an organization

Module 3
Connecting Agents to Digital Ecosystems

1. Construct a use case demonstrating agent-based interaction across integrated tools
2. Write a structured email-style proposal that outlines a specific use case for an AI agent within an organizational context
3. Analyze a business workflow to determine how an AI agent could improve efficiency, reduce costs, or enhance user experience
4. Design an integration approach that specifies how the proposed agent would connect with existing systems, platforms, or application programming interfaces (APIs)
5. Evaluate the potential risks, ethical considerations, and success metrics associated with deploying the proposed AI agent

Module 4
Cybersecurity: Classic Scenarios, Agent Risks, Disinformation, and Systemic

1. Analyze Organizational AI Systems and Workflows to Identify Potential Cybersecurity Risks Using the National Institute of Standards and Technology (NIST) Cybersecurity Framework Categories
2. Evaluate Current Security Practices to Identify Gaps in Access Control, Monitoring, Response, and Recovery Capabilities
3. Develop a Structured AI Risk and Security Plan, Including Stakeholders, Training, and Incident Response Procedures
4. Recommend Actions to Improve Organizational Readiness Across Identify, Protect, Detect, Respond, and Recover Domains
5. Analyze Organizational AI Systems and Workflows to Identify Potential Cybersecurity Risks
6. Evaluate How Accountability Is Defined and Enforced Alongside Security Practices and Governance in AI

Module 5
AI Agents by Business Function

1. Describe the Organizational Context, Including Industry, Organization Type, and Department, Relevant to a Proposed AI-Driven Product Design Initiative
2. Summarize the Current Product Design Workflow Within an Organization to Establish a Baseline for Improvement
3. Select an Appropriate AI Technology for Integration Into a Product Design Process Based on Its Capabilities and Relevance
4. Develop a Structured Plan Outlining How AI Can Be Integrated Into a Product Design Workflow to Improve Efficiency, Effectiveness, or Quality
5. Identify an Appropriate AI Agent Architecture for a Given Organizational Context and Explain Key Trade-Offs
6. Identify Opportunities for AI-Enabled BPO and Describe Their Potential Organizational Impact

Module 6
The Last Mile – From Pilot to Practice

1. Propose Measurable Key Performance Indicators (KPIs) That Evaluate the Effectiveness of an AI System in Relation to Business Outcomes
2. Describe the Organizational Context, Including Sector, Organization Type, and Department, Relevant to an AI Implementation
3. Summarize the Purpose and Functionality of a Proposed AI System Within a Business Workflow
4. Write Three to Five Key Performance Indicators (KPIs) That Measure the Effectiveness of an AI Implementation
5. Evaluate How the Selected KPIs Align With Business Goals and Indicate Whether the AI System Is Achieving Its Intended Outcomes

Module 7
Governance, Compliance, and Agent Testing

1. Identify Applicable Regulatory Frameworks (e.g., GDPR, CCPA, HIPAA) Relevant to a Specific AI Use Case
2. Analyze the Risks Associated With Deploying AI Systems, Including Both Compliance and Operational Risks
3. Apply Appropriate Testing Strategies (e.g., Sandboxing, A/B Testing, Safety Checks) to Evaluate AI System Behavior
4. Develop a Comprehensive AI Governance Plan That Integrates Regulations, Testing, Risk Mitigation, and Documentation Practices
6. Create Guiding Questions That Identify Key Regulatory and Implementation Considerations in Real-World AI Healthcare Scenarios
7. Classify AI Use Cases Using the Risk–Speed Quadrant Framework

Module 8
Ethics and Capstone

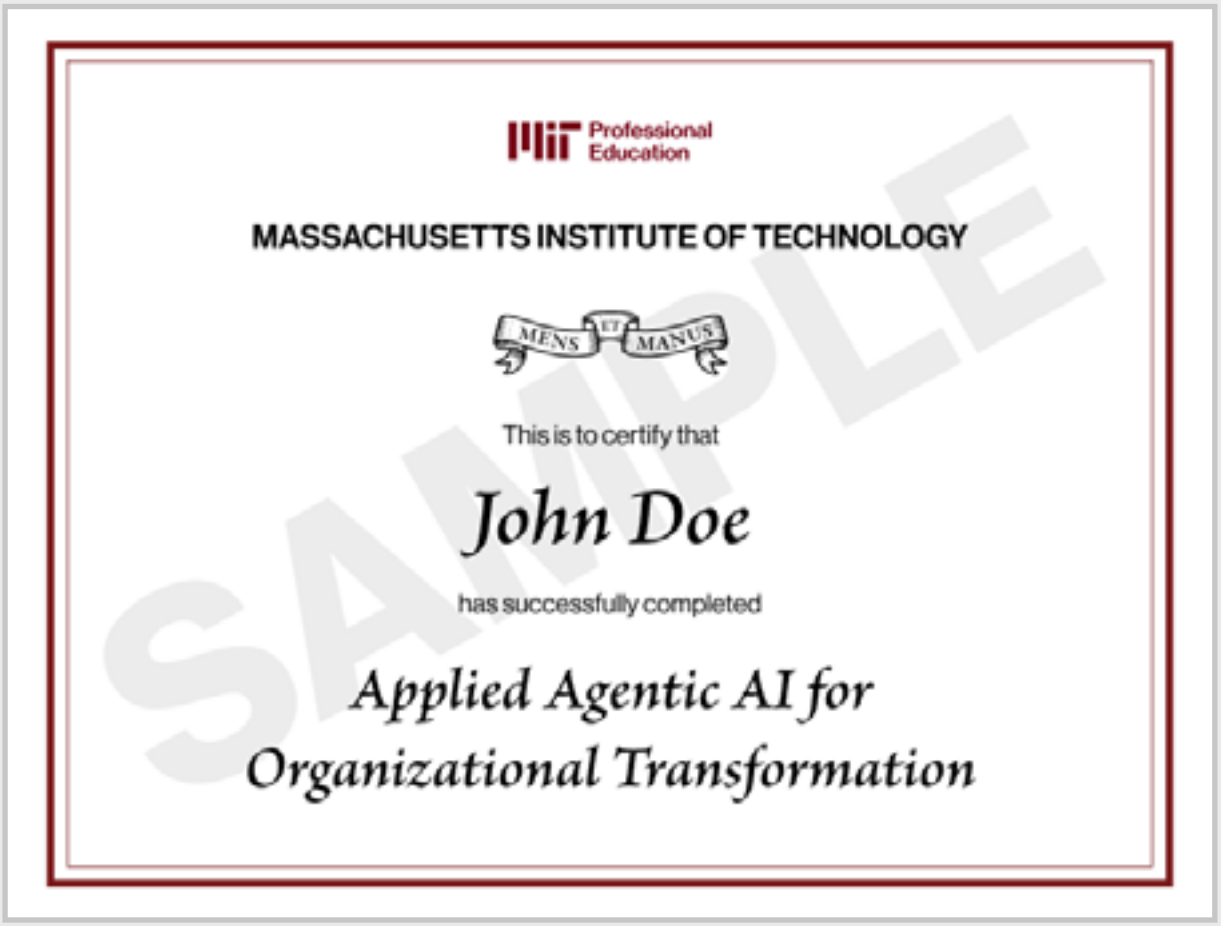
1. Explain How AI Can Be Strategically Integrated Into Organizational Functions to Create Business Value
2. Evaluate the Suitability of AI Technologies for Specific Organizational Use Cases
3. Analyze the Cost, Security, and Operational Implications of AI Adoption
4. Assess the Human and Organizational Factors That Influence Successful AI Implementation
5. Synthesize Course Concepts Into a Structured Approach for Organizational AI Adoption
6. Evaluate Ethical Risks in a Proposed AI System by Identifying a Potential Issue, Assessing Its Business Impact, and Recommending an Appropriate Mitigation Strategy

*Modules and curriculum are subject to change

APPLY NOW

Certificate of Completion

All participants who successfully complete **Applied Agentic AI for Organizational Transformation** will receive an MIT Professional Education Certificate of Completion. In order to obtain CEUs, participants must complete a required CEU accreditation form. CEUs are calculated based on the number of learning hours in each course.



The Continuing Education Unit (CEUs) is defined as 10 contact hours of ongoing learning to indicate the amount of time they have devoted to a non-credit/non-degree professional development program. To understand whether or not these CEUs may be applied toward professional certification, licensing requirements, or other required training or continuing education hours, please consult your training department or licensing authority directly.



Welcome to the gateway

to MIT knowledge and expertise for professionals around the world.

155

Participants from 155 countries

+1600

More than 1,600 online activities

92%

92% of Professional Alumni rated their experience as outstanding

About MIT Professional Education

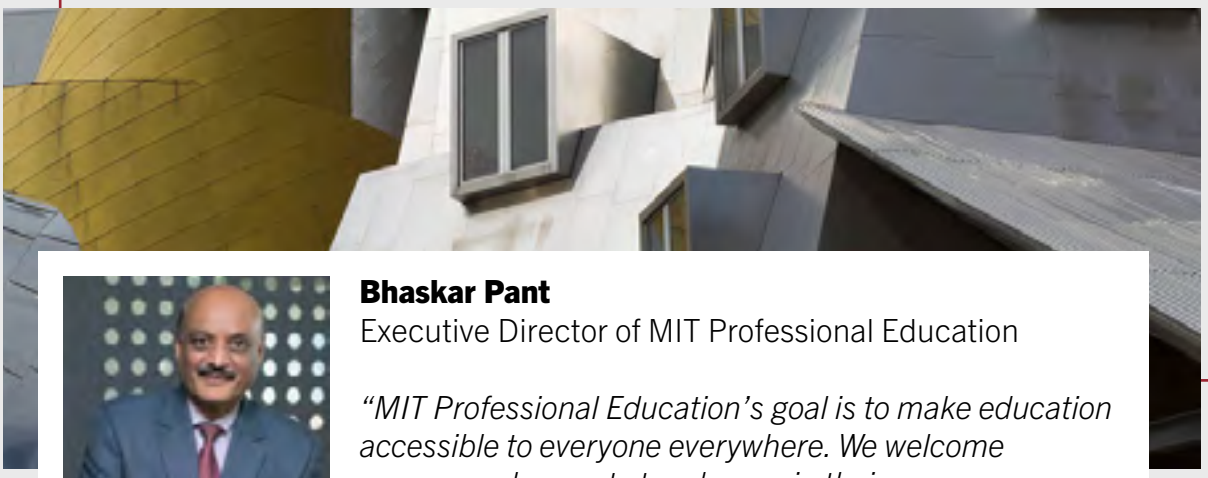
MIT Professional Education fosters the development of innovative leaders equipped to address complex problems globally

MIT Professional Education provides continuing education courses and lifelong learning opportunities for science, engineering, and technology professionals at all levels, from around the world. It is noteworthy that **MIT professors and lecturers lead and teach all MIT Professional Education offerings**. Some of our programs have a long history. Others are relatively new.

Our Digital Plus Programs go beyond online, blending innovative content and delivered using the best of online technology and traditional classroom instruction in a flexible, collaborative learning environment.

MIT Professional Education is central to MIT's vision. It fulfills the mandate to connect practitioner-oriented education with industry and incorporate industry feedback and knowledge into MIT education and research.

The Institute is committed to generating, disseminating, and preserving knowledge while working with others to apply this **knowledge for the benefit of humankind.**





Bhaskar Pant
Executive Director of MIT Professional Education

"MIT Professional Education's goal is to make education accessible to everyone everywhere. We welcome everyone who wants to advance in their career or grow their business to unite and be part of the MIT Professional Education all-inclusive community."

Corporate Courses

At MIT Professional Education, we are committed to accompanying organizations as they aspire to provide their professionals with the ongoing education critical for **success in today's competitive environment**. That's why we offer plans for companies who wish to train their professionals with us.

Which program best suits your organization's needs?

1

Standard Program:
Teams or groups of at least 10 people will obtain special conditions in their registration for any of our programs.

2

Custom Programs
Custom-made to meet the specific needs of each organization and its professionals.

3

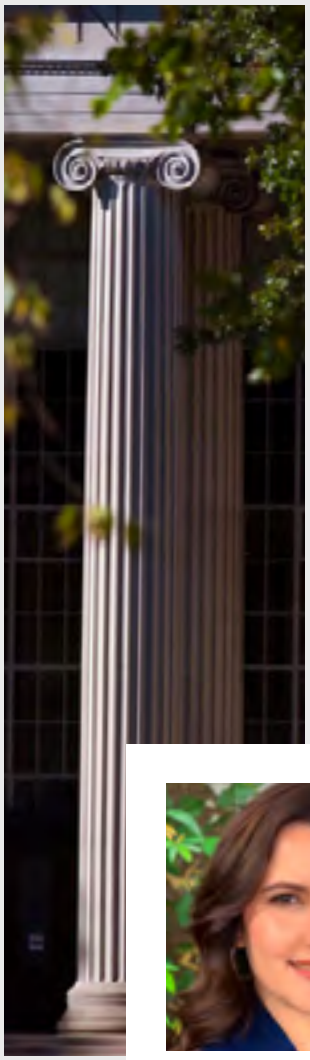
Private Cohorts:
Available in Open-enrollment courses with a minimum number of participants.



The Beyond Online Methodology

MIT Professional Education is revolutionizing the digital learning experience. In fact, we are no longer talking about online-only, but rather an interactive and collaborative learning experience that's digitally powered: a Beyond Online experience.

You will have access to one of the most innovative e-learning platforms that utilizes the latest technology. You will also have an opportunity to meet and work with some of the world's leading subject matter experts in their respective fields.



With MIT Professional Education, you'll have an opportunity to:

- Engage in two live online sessions with MIT instructors per course, and up to eight live sessions with learning facilitators, industry experts, and peers
- Gain practical insights from industry experts and real-world applications
- Fulfill your reskilling and upskilling needs in a fast-changing environment
- Master the technical and human skills necessary for effective leadership
- Meet and network with program participants from around the world



Clara Piloto
Senior Director of Global Programs, Digital Plus Programs and Corporate Programs

"We are currently amid Digital Transformation and Industry 4.0. It is now necessary to understand AI, IoT, cloud, blockchain, and cybersecurity to maintain a competitive edge professionally. Leverage the digital transformation to spark innovation and growth in your

Benefits of Joining the Professional Alumni Community



MIT Professional Education offers a number of benefits for participants who successfully complete our courses:

1. Digital Certificate of Completion and MIT CEUs*.
2. 15% benefit for online programs and short-duration, in-person programs offered by MIT Professional Education.
3. Access to the virtual campus for 6-months after program completion.
4. Exclusive access to the MIT Professional Education Alumni LinkedIn group, connecting you with a global network of fellow MIT Professional Education alumni.

*To obtain CEUs, complete the accreditation confirmation, which is available at the end of the course. CEUs are calculated for each course based on the number of learning hours.

APPLY NOW

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Are you ready to take charge of the latest technologies?

Frequently Asked Questions

Delivered in collaboration with global alumni -excellence or nothing-

For more information, please contact us:

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Website: **https://professionalprograms.mit.edu**

MIT Professional Education is collaborating with online education provider Global Alumni to offer a portfolio of online programs. These programs leverage MIT's leadership in innovation, science, engineering, and technical disciplines developed over years of research, teaching, and practice. Global Alumni collaborates with institutions to manage enrollments (including all payment services and invoicing), technology, and participant support.